17/3/2026

DAY-2

Block-1: Aptitude

1.) A shopkeeper buys a book for ₹400 and sells it for ₹480. Find the profit percentage.

Soln:

$100\% = ₹400$

$x\% = ₹480.$

$$\frac{100}{x} = \frac{\cancel{400}^{5}}{\cancel{480}_{6}}$$

$600 = 5x$

$$x = \frac{\cancel{600}^{120}}{\cancel{5}}$$

$$\boxed{x = 120}$$

Profit $= 120 - 100$

$= 20\%$

2.) A Man buys a watch for ₹1500 and sells it for ₹1200. Find the loss percentage?

Soln:

$100\% = ₹1500$

$x\% = ₹1200$

$$\frac{100}{x} = \frac{\cancel{1500}^{5}}{\cancel{1200}_{4}}$$

$$\frac{\cancel{400}^{80}}{\cancel{5}} = x \Rightarrow \boxed{x = 80}$$

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Loss = 100 − 80

= 20 %

3.) A trader marks a product 20% above the cost price and gives a 10% discount. Find the final profit percentage.

Soln:-

increased = 100% + 20%

= 120%

Discount = 120 − 10% $\frac{120}{10}$

= 120 − 12

= 108

Profit = CP − SP

= 100 − 108

$= \frac{8}{100} \times 100$

Profit = 8%.

4.) If the cost price of 5 pens is ₹100, what is the selling price of each pen to make 25% profit?

Soln

price of 5 pens = ₹100

price of 1 pen = $\frac{100}{5}$ = 20

1 Pen = 20.

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1 Pen profit $= \frac{25}{100} \times 20$

$= ₹5$

5 Pen profit $= 5 \times 5$

$= ₹25$

5.) A product is sold for ₹ 900 with a 10% Loss. Find the cost price

Soln:

Selling price = ₹ 900

Loss % = 10%

100% = 900

Loss = 100 − 10

Selling Price = 90%

$900 = \frac{90}{100} \times x$

$900 \times \frac{10}{9} = x$

$1000 = x$

Cost price = 1000

Block - 2 Verbal Ability.

Reading Passage:

1.) What technologies are influencing software development?

A Cloud computing.

A Artificial Intelligence.

2.) Why are scalable systems important?

Allow companies to handle millions of users simultaneously without performance degradation.

3.) What does "Scalable" mean in software systems?

In software, scalable refers to a system's ability to grow and handle increased demand.

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Grammar Correction:-

1.) He don't like programming.

He doesn't like programming.

2.) She have submitted the assignment.

She has submitted the assignment.

3.) They was discussing the project.

They were discussing the project.

Block - 3 : Communication Practice

Automatic Timetable Generator.

In many colleges, timetable creation is done by manually. It takes Time-consuming. Error-prone (conflicts in faculty/class room and Difficult to manage

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when changes occur. The problem is to design a system that can automatically generate a conflict-free timetable based on faculty availability, subject requirements and Department with Semester constraints. In this project, I used the MERN Stack, which includes MongoDB, Express.js, React.js and Node.js. The system allows the admin to select department and Semester, and with a single click, it generates a timetable automatically.

My role in this project was to develop both frontend and backend, design the database. One of the main challenges I faced was handling scheduling conflicts and ensuring that the timetable is balanced and realistic. Overall, this project helped me to improve my full-stack development skills.

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